

(a) Candidate contract

Correctness-eligible pairs
family-wise fixed hash
amended before sweeps

Identical SFT / RF rule
maximum multiplicity
RTL-hash tie-break

DUT / golden self-check
implementation gate

Frozen image bundle
bitstream + HWH
selectors + manifest

(b) PYNQ-Z2 test harness

Processing system (PS)

Python sweep controller · MMIO · programmable FCLK

load once

Programmable logic (PL): one co-resident image

FCLK

SFT

FIR

rFIR

poly

IIR

med

RF

FIR

rFIR

poly

IIR

med

Counter
stimulus

Echo canary

sel

MUX

Capture
buffer
AXI-lite

capture / reset control

MMIO

(c) Trace validation

Captured traces
fixed clock grid
repeated sweeps

Per-DUT golden
reference stream

Latency alignment
score threshold
canary headroom

Clock boundaries
all paired outcomes

Solid: data / execution path Dashed: control / frozen-reference path